Võ Văn Vĩ
22001515
Buổi 4: Docker

Nhóm lệnh cơ bản

1. Kiểm tra phiên bản Docker

docker --version

```
C:\Users\Johan>docker --version
Docker version 29.2.0, build 0b9d198

C:\Users\Johan>
```

2. Chạy image hello-world để test Docker

docker run hello-world

```
C:\Users\Johan>docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/


C:\Users\Johan>
```

3. Tải image nginx

docker pull nginx

```
C:\Users\Johan>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
5e815e07e569: Pull complete
ec781dee3f47: Pull complete
587e3d84dbb5: Pull complete
3189680c601f: Pull complete
510ddf6557d6: Pull complete
cde7a05ae428: Pull complete
bb3d0aa29654: Pull complete
96a6cfe061e0: Download complete
669e0ab8e7fa: Download complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

C:\Users\Johan>
```

4. Xem các image đã có

docker images

```
C:\Users\Johan>docker images
                                                                                    Info →   U  In Use
IMAGE                              ID             DISK USAGE   CONTENT SIZE   EXTRA
amazon/dynamodb-local:latest       d89f8fcc6b1a        754MB          238MB    U
hello-world:latest                 452a468a4bf9       25.9kB         9.49kB    U
mysql:8.0                          99d774bf02a4       1.08GB          247MB    U
nginx:latest                       7150b3a39203        240MB         65.8MB

C:\Users\Johan>
```

5. Chạy nginx ở chế độ nền

docker run -d nginx

```
C:\Users\Johan>docker run -d nginx
aabb56a2080492bdb0c2db31c05d8fb3f15ceaae3b2183933ec4f2648e9f8c41

C:\Users\Johan>
```

6. Xem container đang chạy

docker ps

```
C:\Users\Johan>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS     NAMES
aabb56a20804   nginx     "/docker-entrypoint.…"   3 minutes ago   Up 3 minutes   80/tcp    awesome_pasteur

C:\Users\Johan>
```

7. Xem tất cả container, kể cả đã dừng

docker ps -a

```
C:\Users\Johan>docker ps -a
CONTAINER ID   IMAGE                          COMMAND                  CREATED         STATUS                     PORTS
    NAMES
aabb56a20804   nginx                          "/docker-entrypoint.…"   3 minutes ago   Up 3 minutes               80/tcp
    awesome_pasteur
74029fcf32ca   hello-world                    "/hello"                 8 minutes ago   Exited (0) 8 minutes ago
    vigorous_leavitt
4c25d5752f72   hello-world                    "/hello"                 8 minutes ago   Exited (0) 8 minutes ago
    agitated_vaughan
c50ca4beea25   amazon/dynamodb-local:latest   "java -jar DynamoDBL…"   6 days ago      Exited (143) 3 days ago
    dynamodb-local
656bc233a411   mysql:8.0                      "docker-entrypoint.s…"   6 weeks ago     Exited (0) 6 weeks ago
    spring_boot_mysql

C:\Users\Johan>
```

8. Xem log của container

docker logs <container_id>

```
C:\Users\Johan>docker logs aabb56a20804
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/26 11:43:44 [notice] 1#1: using the "epoll" event method
2026/03/26 11:43:44 [notice] 1#1: nginx/1.29.7
2026/03/26 11:43:44 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/26 11:43:44 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/26 11:43:44 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/26 11:43:44 [notice] 1#1: start worker processes
2026/03/26 11:43:44 [notice] 1#1: start worker process 29
2026/03/26 11:43:44 [notice] 1#1: start worker process 30
2026/03/26 11:43:44 [notice] 1#1: start worker process 31
2026/03/26 11:43:44 [notice] 1#1: start worker process 32
2026/03/26 11:43:44 [notice] 1#1: start worker process 33
2026/03/26 11:43:44 [notice] 1#1: start worker process 34
2026/03/26 11:43:44 [notice] 1#1: start worker process 35
2026/03/26 11:43:44 [notice] 1#1: start worker process 36
2026/03/26 11:43:44 [notice] 1#1: start worker process 37
2026/03/26 11:43:44 [notice] 1#1: start worker process 38
2026/03/26 11:43:44 [notice] 1#1: start worker process 39
2026/03/26 11:43:44 [notice] 1#1: start worker process 40

C:\Users\Johan>
```

Ví dụ:

docker logs aabb56a20804

9. Vào bên trong container bằng shell

docker exec -it <container_id> /bin/sh

```
C:\Users\Johan>docker exec -it aabb56a20804 /bin/sh
#
```

Ví dụ:

docker exec -it aabb56a20804 /bin/sh

10. Dừng container

docker stop <container_id>
docker stop aabb56a20804

```
C:\Users\Johan>docker stop aabb56a20804
aabb56a20804

C:\Users\Johan>
```

11. Khởi động lại container

docker restart <container_id>

docker restart aabb56a20804

```
C:\Users\Johan>docker restart aabb56a20804
aabb56a20804

C:\Users\Johan>
```

12. Xóa container

docker rm <container_id>
docker rm aabb56a20804

```
C:\Users\Johan>docker stop aabb56a20804
aabb56a20804

C:\Users\Johan>docker rm aabb56a20804
aabb56a20804

C:\Users\Johan>
```

Nhóm dọn dẹp

13. Xóa các container đã dừng

docker container prune

```
C:\Users\Johan>docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
74029fcf32cab29b898bd501ea4db6c2331d9c9c024b41a8e129fecc4490ecfb
4c25d5752f7214b465e573ad8c5c425449f7c118c670ba1de4dcb68acacae748
c50ca4beea2590c5bd4c516aace9bee9b22692b63d8e1789c60c50999f852a0d
656bc233a4111e83dcb9d36ddac7ed1488a931e6cd2e3eb7792e5e8aa9711f8f

Total reclaimed space: 966.7kB

C:\Users\Johan>
```

14. Xóa image theo ID

docker rmi <image_id>
docker rmi 452a468a4bf9

```
C:\Users\Johan>docker images
                                                                                    Info →   U  In Use
IMAGE                                   ID             DISK USAGE   CONTENT SIZE   EXTRA
amazon/dynamodb-local:latest            d89f8fcc6b1a        754MB          238MB
hello-world:latest                      452a468a4bf9       25.9kB         9.49kB
mysql:8.0                               99d774bf02a4       1.08GB          247MB
nginx:latest                            7150b3a39203        240MB         65.8MB

C:\Users\Johan>docker rmi 452a468a4bf9
Untagged: hello-world:latest
Deleted: sha256:452a468a4bf985040037cb6d5392410206e47db9bf5b7278d281f94d1c2d0931

C:\Users\Johan>
```

15. Xóa toàn bộ image không dùng

docker image prune -a

```
C:\Users\Johan>docker image prune -a
WARNING! This will remove all images without at least one container associated to them.
Are you sure you want to continue? [y/N] y
Deleted Images:
untagged: amazon/dynamodb-local:latest
deleted: sha256:d89f8fcc6b1a39cb35976c248ed42a28c66ae00dc043099210f5571e42648ab4
deleted: sha256:6b954943dc44e9f0898a75e17192280f558550cf7651cf1b6546af9c55975526
deleted: sha256:d6843ec545dd2b8ee0584840d2e6f929d7f83fbabf7ae6aa50a132da2adaed59
untagged: mysql:8.0
deleted: sha256:99d774bf02a48a1bb1c599920d2571946d31e5940b854b02737d5e95c184358f
deleted: sha256:1681328bca475f89da97cf8b874aa8a6ef3f839bde63e94d65e0f3bb0a62c94f
deleted: sha256:99aa9dbc770f0d94952937ba365e952bd63760c0c858c55c5171309544783734
untagged: nginx:latest
deleted: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
deleted: sha256:c3fe1eeae810f4a585961f17339c93f0fb1c7c8d5c02c9181814f52bdd51961c
deleted: sha256:b656f56499226a9982c37382327226952086a9550d6abd642873cb9089b922bd

Total reclaimed space: 550.8MB

C:\Users\Johan>
```

Nhóm port, volume, inspect

16. Chạy nginx và map cổng 8080 máy thật vào 80 của container

docker run -d -p 8080:80 nginx

Sau đó mở trình duyệt:

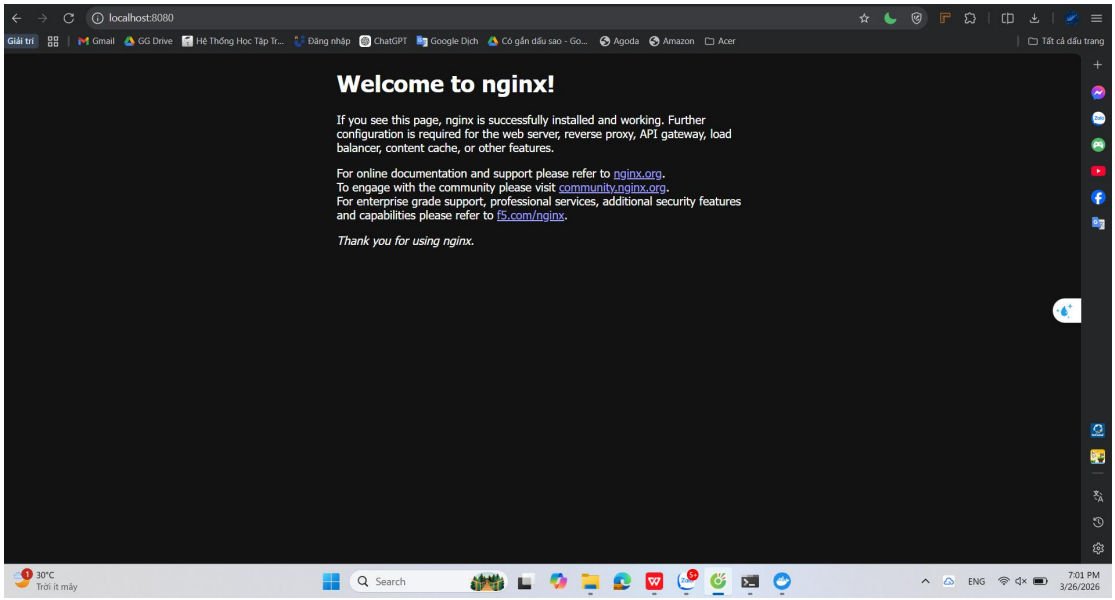
http://localhost:8080

```
C:\Users\Johan>docker run -d -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
5e815e07e569: Pull complete
3189680c601f: Pull complete
ec781dee3f47: Pull complete
cde7a05ae428: Pull complete
bb3d0aa29654: Pull complete
510ddf6557d6: Pull complete
587e3d84dbb5: Pull complete
669e0ab8e7fa: Download complete
96a6cfe061e0: Download complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
247a7876a00d3e6b778fddebb2980c381a48d2277c8b1b29525c9b9f46a9a633

C:\Users\Johan>docker run -d -p 8080:80 nginx
44796d325bbdcc7d96d09a488f4dd314950d2e18e0f43135b6d06b442df75d66
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivi
ty on endpoint elated_swirles (a492ab22a63e3ab4ccf018baef310966b21c5f03130bc7c43e59da01bd8126e9): Bind for 0.0.0.0:8080
failed: port is already allocated

Run 'docker run --help' for more information

C:\Users\Johan>
```



17. Xem thông tin chi tiết container

```
docker inspect <container_id>
docker inspect 247a7876a00d
```

```
C:\Users\Johan>docker inspect 247a7876a00d
[
    {
        "Id": "247a7876a00d3e6b778fddebb2980c381a48d2277c8b1b29525c9b9f46a9a633",
        "Created": "2026-03-26T12:00:11.390394276Z",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 914,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-03-26T12:00:13.123047764Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18",
        "ResolvConfPath": "/var/lib/docker/containers/247a7876a00d3e6b778fddebb2980c381a48d2277c8b1b29525c9b9f46a9a633/r
esolv.conf",
        "HostnamePath": "/var/lib/docker/containers/247a7876a00d3e6b778fddebb2980c381a48d2277c8b1b29525c9b9f46a9a633/hos
tname",
```

18. Gắn volume thư mục máy thật vào containerax

```
docker run -d -v mydata:/data nginx
```

```
C:\Users\Johan>docker run -d -v mydata:/data nginx
63fa94f0a378a5922643297e3015c5f35a19a933ff8f7814136d9b0e48efef9c

C:\Users\Johan>
```

19. Xem danh sách volume

```
docker volume ls
```

```
C:\Users\Johan>docker volume ls
DRIVER    VOLUME NAME
local     demo-spring_db_data
local     mydata

C:\Users\Johan>
```

20. Xóa volume không dùng

docker volume prune

```
C:\Users\Johan>docker volume ls
DRIVER    VOLUME NAME
local     demo-spring_db_data
local     mydata

C:\Users\Johan>docker volume prune
WARNING! This will remove anonymous local volumes not used by at least one container.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B

C:\Users\Johan>
```

Nhóm đặt tên container

21. Chạy nginx và đặt tên container

docker run -d --name my_nginx nginx

```
C:\Users\Johan>docker run -d --name my_nginx nginx
8c029f060bf86168aa29fa4c86f2e4d5fe36cf6256dbeddeb0e269277d11bf70

C:\Users\Johan>
```

22. Xem tài nguyên container

docker stats

```
CONTAINER ID   NAME              CPU %     MEM USAGE / LIMIT     MEM %     NET I/O           BLOCK I/O         PIDS
8c029f060bf8   my_nginx          0.00%     10.52MiB / 7.437GiB   0.14%     872B / 126B       0B / 12.3kB       13
63fa94f0a378   admiring_wescoff  0.00%     12.76MiB / 7.437GiB   0.17%     998B / 126B       4.05MB / 12.3kB   13
247a7876a00d   gifted_torvalds   0.00%     10.49MiB / 7.437GiB   0.14%     4.15kB / 3.28kB   12.3kB / 12.3kB   13
```

Nhóm network

23. Xem các network

docker network ls

```
C:\Users\Johan>docker network ls
NETWORK ID     NAME                           DRIVER    SCOPE
878af021f97b   bridge                         bridge    local
23e8bb987737   demo-spring_default            bridge    local
4abf8d7831cc   host                           host      local
3c59d68c4d3f   lab6-product-manager_default   bridge    local
fb630eafcbf5   none                           null      local

C:\Users\Johan>
```

24. Tạo network mới

docker network create my_network

```
C:\Users\Johan>docker network create my_network
2203bb9858c6acfc5ea334a9b753daaaeaf3ae0e03ef19bbf6cb1ac3efb44d03

C:\Users\Johan>docker network ls
NETWORK ID     NAME                           DRIVER    SCOPE
878af021f97b   bridge                         bridge    local
23e8bb987737   demo-spring_default            bridge    local
4abf8d7831cc   host                           host      local
3c59d68c4d3f   lab6-product-manager_default   bridge    local
2203bb9858c6   my_network                     bridge    local
fb630eafcbf5   none                           null      local

C:\Users\Johan>
```

25. Chạy container trong network vừa tạo

docker run -d --network my_network --name my_container nginx

```
C:\Users\Johan>docker run -d --network my_network --name my_container nginx
e966a85561b468847d8ae5bab51ee7ed533db4a721d4537220825cd3daf1fb39

C:\Users\Johan>
```

26. Kết nối container có sẵn vào network

docker network connect my_network my_nginx

```
C:\Users\Johan>docker network connect my_network my_nginx

C:\Users\Johan>
```

Biến môi trường

27. Chạy container với biến môi trường

docker run -d -e MY_ENV=hello_world nginx

```
C:\Users\Johan>docker run -d -e MY_ENV=hello_world nginx
e583c7eb5e0dc0f6e36f2abdb29cb76b2fdd40f4bb96f874059a8d16f6c06c6f

C:\Users\Johan>
```

Xem log theo tên container

28. Xem log của container tên my_nginx

```
docker logs -f my_nginx
```

```
C:\Users\Johan>docker logs -f my_nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/26 12:15:23 [notice] 1#1: using the "epoll" event method
2026/03/26 12:15:23 [notice] 1#1: nginx/1.29.7
2026/03/26 12:15:23 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/26 12:15:23 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/26 12:15:23 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/26 12:15:23 [notice] 1#1: start worker processes
2026/03/26 12:15:23 [notice] 1#1: start worker process 29
2026/03/26 12:15:23 [notice] 1#1: start worker process 30
2026/03/26 12:15:23 [notice] 1#1: start worker process 31
2026/03/26 12:15:23 [notice] 1#1: start worker process 32
2026/03/26 12:15:23 [notice] 1#1: start worker process 33
2026/03/26 12:15:23 [notice] 1#1: start worker process 34
2026/03/26 12:15:23 [notice] 1#1: start worker process 35
2026/03/26 12:15:23 [notice] 1#1: start worker process 36
2026/03/26 12:15:23 [notice] 1#1: start worker process 37
2026/03/26 12:15:23 [notice] 1#1: start worker process 38
2026/03/26 12:15:23 [notice] 1#1: start worker process 39
2026/03/26 12:15:23 [notice] 1#1: start worker process 40
```

---

3) Cách chạy phần Dockerfile

Ba dòng này không gõ chung một chỗ với lệnh Docker, mà cần tạo file trước:

```
FROM nginx
COPY index.html /usr/share/nginx/html/index.html
```

Bước 1: Tạo thư mục dự án

Ví dụ tạo thư mục:

```
docker-nginx-demo
```

Bước 2: Trong thư mục đó tạo 2 file

File 1: `Dockerfile`

```
FROM nginx
COPY index.html /usr/share/nginx/html/index.html
```

File 2: `index.html`

```
<!DOCTYPE html>
<html>
<head>
    <title>My Nginx</title>
</head>
<body>
    <h1>Hello Docker Nginx</h1>
</body>
</html>
```

Bước 3: Mở terminal tại đúng thư mục đó

Ví dụ:

cd đường_dẫn_đến_thư_mục/docker-nginx-demo

Bước 4: Build image

docker build -t my_nginx_image .

Lưu ý dấu . ở cuối, nghĩa là build từ thư mục hiện tại.

```
C:\Users\Johan\Desktop\TKKTPM\Buoi4_Docker\docker-nginx-demo>docker build -t my_nginx_image .
[+] Building 1.6s (7/7) FINISHED                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                       0.1s
 => => transferring dockerfile: 97B                                                        0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                            0.5s
 => [internal] load .dockerignore                                                          0.1s
 => => transferring context: 2B                                                            0.0s
 => [internal] load build context                                                          0.1s
 => => transferring context: 167B                                                          0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf  0.2s
 => => resolve docker.io/library/nginx:latest@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf  0.1s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                 0.1s
 => exporting to image                                                                     0.4s
 => => exporting layers                                                                    0.1s
 => => exporting manifest sha256:c43fbd90bd0eeac15eb45abe5c9bb66371b58adc6b44a5c8c0a4fe9d4b1  0.0s
 => => exporting config sha256:f3ba11cb6eec752e8fee6de9e4151b798f30d9be481e7105112ef59f513a5  0.0s
 => => exporting attestation manifest sha256:33e677ac88274b32be16dd0a98bd4b65044a80ad977db4c  0.0s
 => => exporting manifest list sha256:1835545f796dc04673f2f4e3e9bdd71d96b2c2ee7d931768fce25a  0.0s
 => => naming to docker.io/library/my_nginx_image:latest                                   0.0s
 => => unpacking to docker.io/library/my_nginx_image:latest                                0.1s

C:\Users\Johan\Desktop\TKKTPM\Buoi4_Docker\docker-nginx-demo>
```

Bước 5: Chạy image vừa build

Trong ảnh của bạn dòng 31 đang là:

```
docker run -d -p 8080:80 my_nginx_image
```

Mình sửa lại cổng cho đúng kiểu Nginx:

```
docker run -d -p 8080:80 my_nginx_image
```

```
C:\Users\Johan\Desktop\TKKTPM\Buoi4_Docker\docker-nginx-demo>docker run -d -p 8080:80 my_nginx_imag
e
c13341dd15ff4b6a519e4b5dce6df3a46bde1738c50ce2c3b9d2596e7ce08f81
docker: Error response from daemon: failed to set up container networking: driver failed programmin
g external connectivity on endpoint competent_hodgkin (91c70a9eced8fca4d92221a0387b5a157530c546902f
9de1d45a434915f891ab): Bind for 0.0.0.0:8080 failed: port is already allocated

Run 'docker run --help' for more information

C:\Users\Johan\Desktop\TKKTPM\Buoi4_Docker\docker-nginx-demo>
```

Sau đó mở:

```
http://localhost:8080
```